Residual Output — Line-by-Line Easy Calculation

This PDF explains Predicted Sales, Residuals, and Standard Residuals for all 15 observations.

1. Start with the regression equation

From the regression analysis:

■ = 74.55655351 + 3.018572081X

where X = Advertising Spend and ■ = Predicted Sales.

2. Predicted Sales calculation

For every observation, use:

Predicted Sales (■) = Intercept + (Slope × Advertising Spend)

■ = 74.55655351 + (3.018572081 × X)

Example — Observation 1

X = 10, Actual Sales = 105

■ = 74.55655351 + (3.018572081 × 10)

■ = 74.55655351 + 30.18572081

■ = 104.7422743

3. Residual calculation

Residual means the difference between Actual Sales and Predicted Sales.

Formula: Residual = Actual Sales – Predicted Sales

Observation 1:

Residual = 105 – 104.7422743 = 0.257725681

4. Standard Residual calculation

A standard residual puts the residual on a standardized scale. For this Excel output, it is calculated using the regression standard error.

Formula: Standard Residual = Residual ÷ Standard Error

Standard Error = √MSE

MSE = 0.640701664

Standard Error = $\sqrt{0.640701664} \approx 0.80043842$

Observation 1:

Standard Residual = $0.257725681 \div 0.80043842$

≈ 0.322

Important note: the supplied Excel Standard Residual values use Excel's residual-output standardization calculation. The table below reproduces the supplied output and explains the basic calculation. Small differences can occur because Excel's standard residual uses leverage-adjusted standardization in its regression output.

5. All 15 observations — line by line

For each row: (1) calculate ■, (2) calculate Residual = Y – ■, and (3) standardize the residual using Excel's residual-output method.

Obs.	X	Actual Y	Predicted Sales ■	Residual Y–■	Standard Residual
1	10	105	104.7422743	0.257725680	0.321980647
2	12	112	110.7794185	1.220581518	1.524891219
3	15	120	119.8351347	0.164865275	0.205968718
4	18	128	128.8908510	-0.890850968	-1.112953784
5	20	135	134.9279951	0.072004870	0.089956789
6	22	141	140.9651393	0.034860708	0.043552017
7	25	150	150.0208555	-0.020855535	-0.026055140
8	28	158	159.0765718	-1.076571778	-1.344977641
9	30	165	165.1137159	-0.113715940	-0.142067069
10	32	171	171.1508601	-0.150860102	-0.188471840
11	35	180	180.2065763	-0.206576345	-0.258078998
12	38	188	189.2622926	-1.262292588	-1.577001499
13	40	195	195.2994367	-0.299436750	-0.374090926
14	42	202	201.3365809	0.663419088	0.828819646
15	45	212	210.3922972	1.607702845	2.008527833

6. Detailed examples from different rows

Observation 2

X = 12, Actual Y = 112

Predicted Sales = 74.55655351 + (3.018572081 × 12) = 110.7794185

Residual = 112 – 110.7794185 = 1.220581518

Standard Residual (basic residual ÷ 0.80043842) = 1.524891219

Observation 4

X = 18, Actual Y = 128

Predicted Sales = 74.55655351 + (3.018572081 × 18) = 128.8908510

Residual = 128 – 128.8908510 = -0.890850968

Standard Residual (basic residual ÷ 0.80043842) = -1.112953784

Observation 8

X = 28, Actual Y = 158

Predicted Sales = 74.55655351 + (3.018572081 × 28) = 159.0765718

Residual = 158 – 159.0765718 = -1.076571778

Standard Residual (basic residual ÷ 0.80043842) = -1.344977641

Observation 12

X = 38, Actual Y = 188

Predicted Sales = 74.55655351 + (3.018572081 × 38) = 189.2622926

Residual = 188 – 189.2622926 = -1.262292588

Standard Residual (basic residual ÷ 0.80043842) = -1.577001499

Observation 15

X = 45, Actual Y = 212

Predicted Sales = 74.55655351 + (3.018572081 × 45) = 210.3922972

Residual = 212 – 210.3922972 = 1.607702845

Standard Residual (basic residual ÷ 0.80043842) = 2.008527833

7. What the residual sign means

Positive residual:

Actual Sales > Predicted Sales. The model predicted too low.

Negative residual:

Actual Sales < Predicted Sales. The model predicted too high.

Residual close to 0:

Prediction is very close to the actual Sales value.

8. Understand your first few rows

Observation	Actual	Predicted	Residual	Meaning
1	105	104.7423	+0.2577	Actual is slightly higher
2	112	110.7794	+1.2206	Actual is higher
3	120	119.8351	+0.1649	Very close
4	128	128.8909	–0.8909	Actual is lower

9. Easy memory method

Step 1 → Put X into the regression equation to get Predicted Sales.

Step 2 → Actual Sales – Predicted Sales = Residual.

Step 3 → Standardize the residual to obtain the Standard Residual.

Positive residual = model predicted too low.

Negative residual = model predicted too high.

Residual near 0 = prediction is close to actual.

10. Final formulas

Predicted Sales: ■ = 74.55655351 + 3.018572081X

Residual: e = Y – ■

Standard residual: Excel uses a standardized residual calculation based on the residual and its estimated standard deviation, including leverage.